

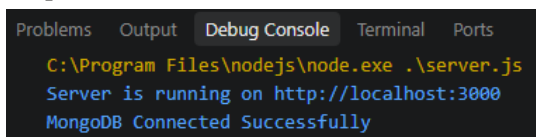
Practical – VIII

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Class: TYIT	Batch: 1
Date of Assignment: 10/07/2026	Date/Time of Submission: 10/07/2026

a. Create a simple REST API using Express

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB") .then(() => {
  console.log("MongoDB Connected Successfully"); })
.catch((error) => {
  console.log("MongoDB Connection Failed");
  console.log(error);
});
// Simple REST API
app.get("/", (req, res) => {
  res.send("Student Management API is Running"); });
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:\${PORT}`);
});
```

Output:



```
C:\Program Files\nodejs\node.exe .\server.js
Server is running on http://localhost:3000
MongoDB Connected Successfully
```

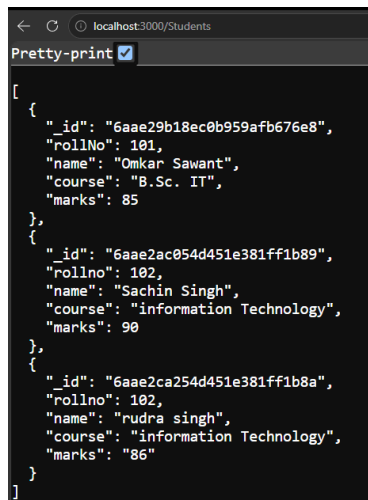


b. Implement a GET API to retrieve data.

Add following code practical (a) code before `app.listen()`

```
app.get("/students", async (req, res) => {
  try {
    const students = await mongoose.connection.db
      .collection("students")
      .find()
      .toArray();
    res.json(students);
  }
  catch (error) {
    res.status(500).json({
      message: "Error fetching students"
    });
  }
});
```

```
});  
}  
});
```



c. Fetch data from the API in a React application.

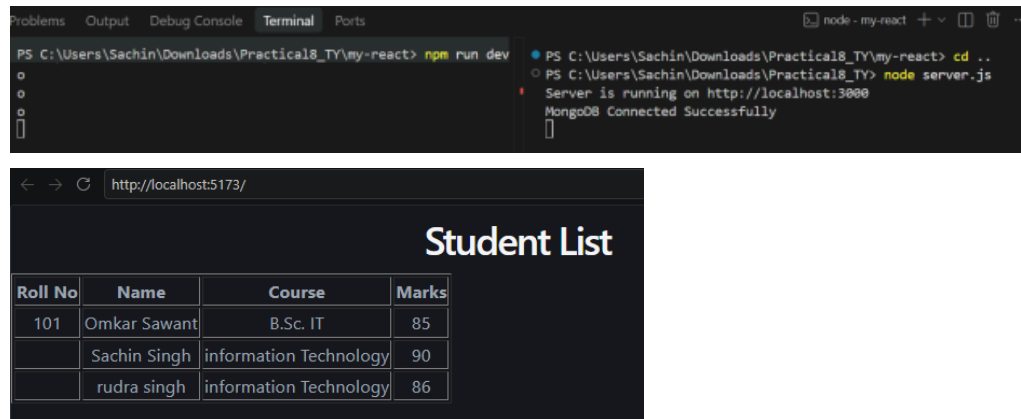
Add following code to App.jsx

```
import { useEffect, useState } from "react";  
import axios from "axios";  
function App() {  
  const [students, setStudents] = useState([]);  
  useEffect(() => { axios.get("http://localhost:3000/students") .then((response) => {setStudents(response.data);})  
    .catch((error) => {console.log(error);}); }, []);  
  return (  
    <div>  
      <h1>Student List</h1>  
      <table border="1">  
        <thead>  
          <tr>  
            <th>Roll No</th>  
            <th>Name</th>  
            <th>Course</th>  
            <th>Marks</th>  
          </tr>  
        </thead>  
        <tbody>  
          {students.map((student) => (  
            <tr key={student._id}>  
              <td>{student.rollNo}</td>  
              <td>{student.name}</td>  
              <td>{student.course}</td>  
              <td>{student.marks}</td>  
            </tr>  
          ))}  
        </tbody>  
      </table>  
    </div>  
  );  
}
```

export default App;

Split terminal & start server in that terminal. To start execute server.js using command node server.js.

Then in 1st terminal execute react application using npm run dev



d. Send data from React to the server using a POST request.

1) Replace App.jsx code

```
import { useEffect, useState } from "react";
import axios from "axios";

function App() {
  const [students, setStudents] = useState([]);
  const [rollNo, setRollNo] = useState("");
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
  const [marks, setMarks] = useState("");

  // Get students
  const getStudents = () => {
    axios.get("http://localhost:3000/students")
      .then((response) => {
        setStudents(response.data);
      })
      .catch((error) => {
        console.log(error);
      });
  };

  useEffect(() => {
    getStudents();
  }, []);

  // Add student
  const addStudent = () => {
    const newStudent = {
      rollNo: Number(rollNo),
      name: name,
      course: course,
      marks: Number(marks)
    };

    axios.post("http://localhost:3000/students", newStudent)
      .then((response) => {
        alert(response.data.message);
        setRollNo("");
        setName("");
        setCourse("");
        setMarks("");
        getStudents();
      })
      .catch((error) => {
        console.log(error);
      });
  };

  return (
    <div>
      <h1>Student Management System</h1>

```

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```
<h2>Add Student</h2>
<input
  type="number"
  placeholder="Roll No"
  value={rollNo}
  onChange={(e) => setRollNo(e.target.value)}
/>
<br /><br />
<input
  type="text"
  placeholder="Student Name"
  value={name}
  onChange={(e) => setName(e.target.value)}
/>
<br /><br />

<input
  type="text"
  placeholder="Course"
  value={course}
  onChange={(e) => setCourse(e.target.value)}
/>
<br /><br />
<input
  type="number"
  placeholder="Marks"
  value={marks}
  onChange={(e) => setMarks(e.target.value)}
/>
<br /><br />
<button onClick={addStudent}>
  Add Student
</button>
<hr />
<h2>Student List</h2>
<table border="1">
  <thead>
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
    </tr>
  </thead>
  <tbody>
    {students.map((student) => (
      <tr key={student._id}>
        <td>{student.rollNo}</td>
        <td>{student.name}</td>
        <td>{student.course}</td>
        <td>{student.marks}</td>
      </tr>
    ))}
  </tbody>
</table>
```

```
</div>
```

```
);  
}
```

```
export default App;
```

2) Add post method in server.js file above app.listen()

```
app.post("/students", async (req, res) => {  
  try {  
    const student = req.body;  
    await mongoose.connection.db  
      .collection("students")  
      .insertOne(student);  
    res.json({  
      message: "Student added successfully",  
      student: student  
    });  
  }  
  catch (error) {  
    res.status(500).json({  
      message: "Error adding student"  
    });  
  }  
});
```

3) Now 1st execute server.js then app.jsx in different terminals.

Output -

The screenshot shows a web browser at <http://localhost:5173/> displaying a web application. The application has two main sections: "Add Student" and "Student List".

The "Add Student" section contains four input fields: "Roll No", "Student Name", "Course", and "Marks", followed by an "Add Student" button.

The "Student List" section displays a table with the following data:

Roll No	Name	Course	Marks
101	Omkar Sawant	B.Sc. IT	85
	Sachin Singh	information Technology	90
	rudra singh	information Technology	86

Below the browser window, the VS Code terminal shows the command prompt and the output of running the application. The terminal shows the following commands and output:

```
PS C:\Users\itlab\Desktop\mern_practical\my-react> npm run dev  
→ press h + enter to show help  
PS C:\Users\itlab\Desktop\mern_practical\my-react> npm run dev  
run dev  
my-react@0.0.0 dev  
vite  
VITE v8.3.0 ready in 185 ms  
→ Local: http://localhost:5173/  
→ Network: use --host to expose
```

The terminal also shows the output of running the server.js file, which includes the message "MongoDB Connected Successfully" and "Server is running on http://localhost:3000".